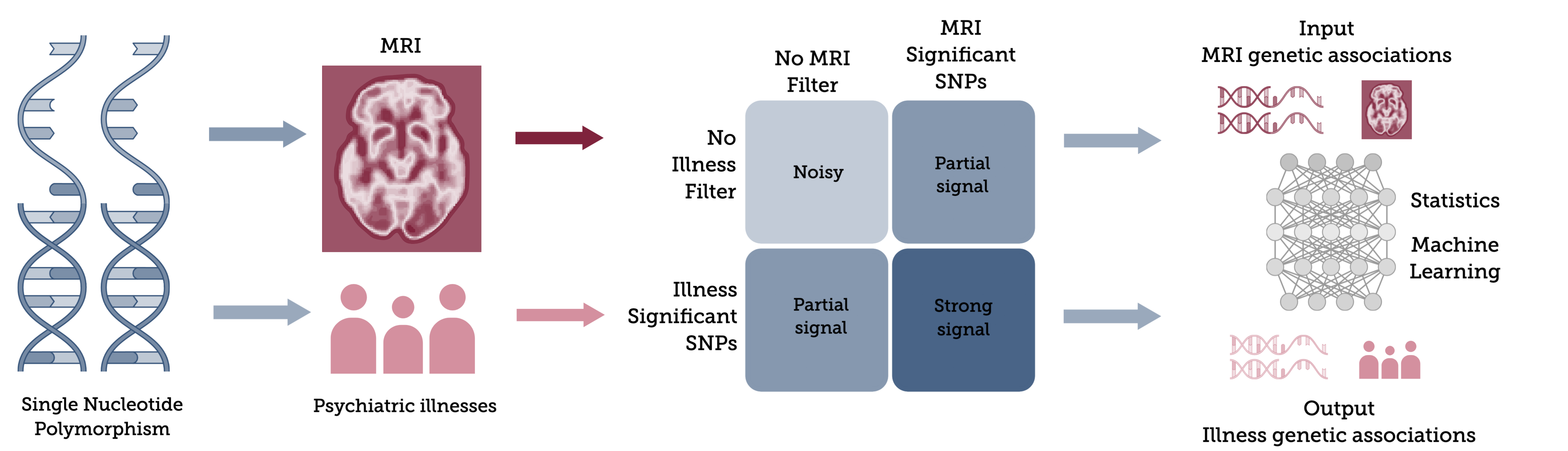
Can SNPs effect on brain structure predict SNPs effect on psychiatric illness ?

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Genetic Variants | Phenotyping for GWAS | 2x2 Experiment Grid | Prediction with Machine Learning



Single Nucleotide Polymorphism
MRI
Psychiatric illnesses
No MRI Filter
MRI Significant SNPs
No Illness Filter
Illness Significant SNPs
Noisy
Partial signal
Partial signal
Strong signal
Input
MRI genetic associations
Statistics
Machine Learning
Output
Illness genetic associations

Dataset

	MRI Genetic Associations				Psychiatric Illnesses Genetic Associations					
SNP	Volume_left	Volume_right	..	Area_right	Schizophrenia	Altzheimers	Depression	Bipolar	ADHD	OCD
rs14563	5.3421	2.4542	0.3432	5.3421	2.4542	0.3432	5.3421	2.4542	0.3432	0.3432
rs4678	-1.2343	-4.3943	1.2321	-1.2343	-4.3943	1.2321	-1.2343	-4.3943	1.2321	1.2321
rs89123	0.3432	-1.2384	-0.8349	0.3432	-1.2384	-0.8349	0.3432	-1.2384	-0.8349	-0.8349

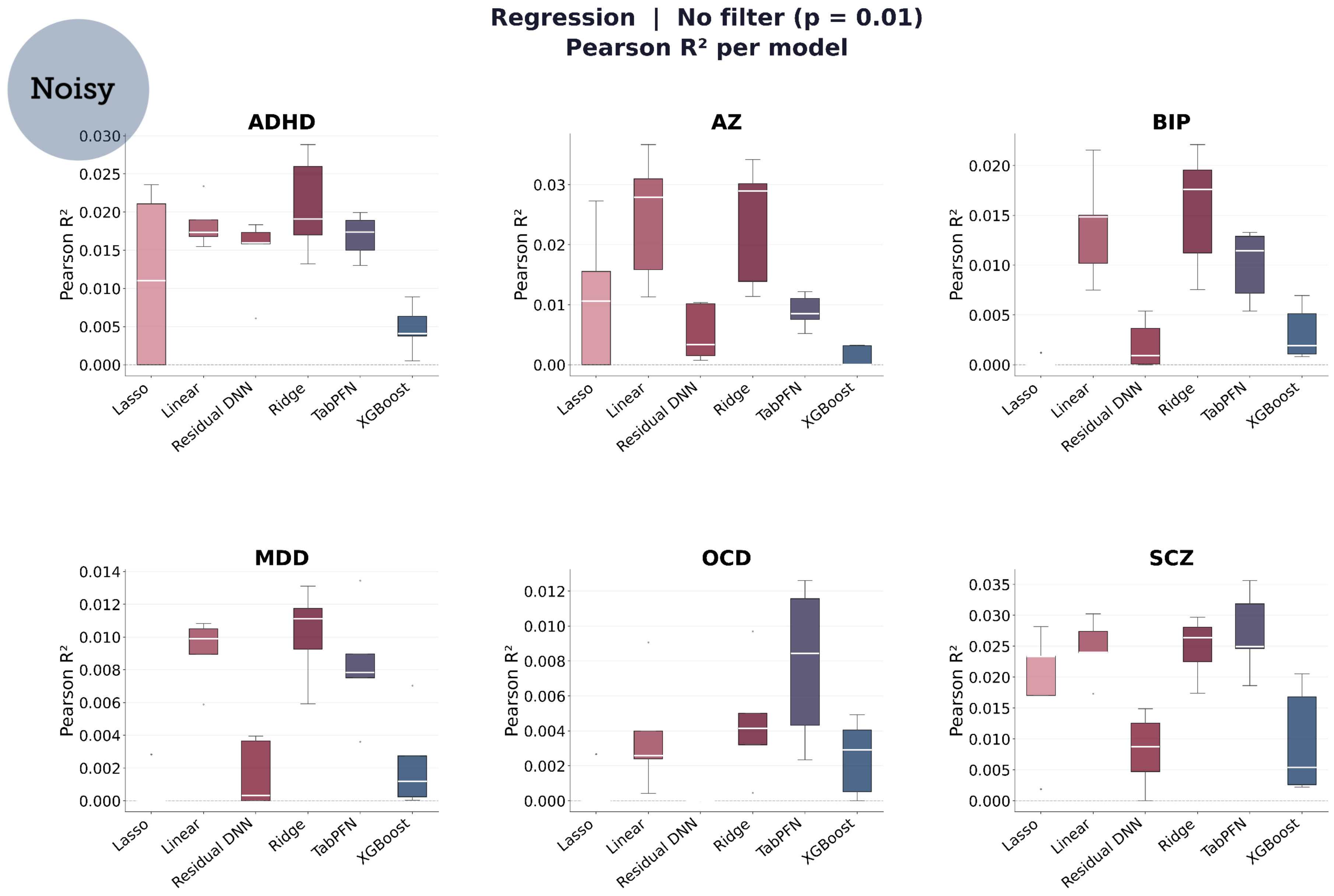
Results

1. **Significance of Genetic Illness-Association Dictate performance**
2. **Non-lienar methods do not outperform linear methods**

Machine Learning Models

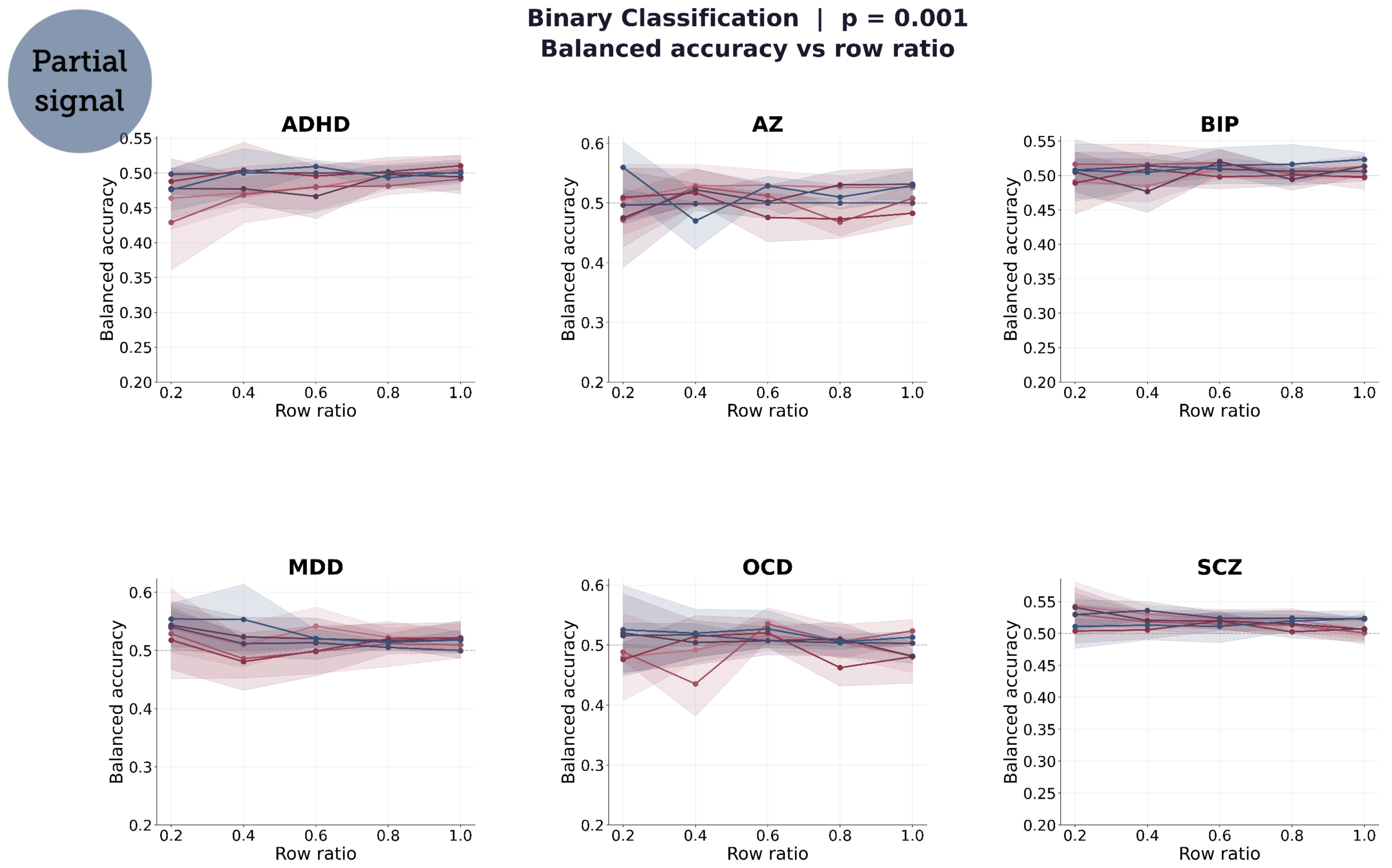
Lasso | Lasso Logistic | Linear | Residual DNN | Ridge | Ridge Logistic | TabPFN | XGBoost

Regression | No filter (p = 0.01)
Pearson R² per model



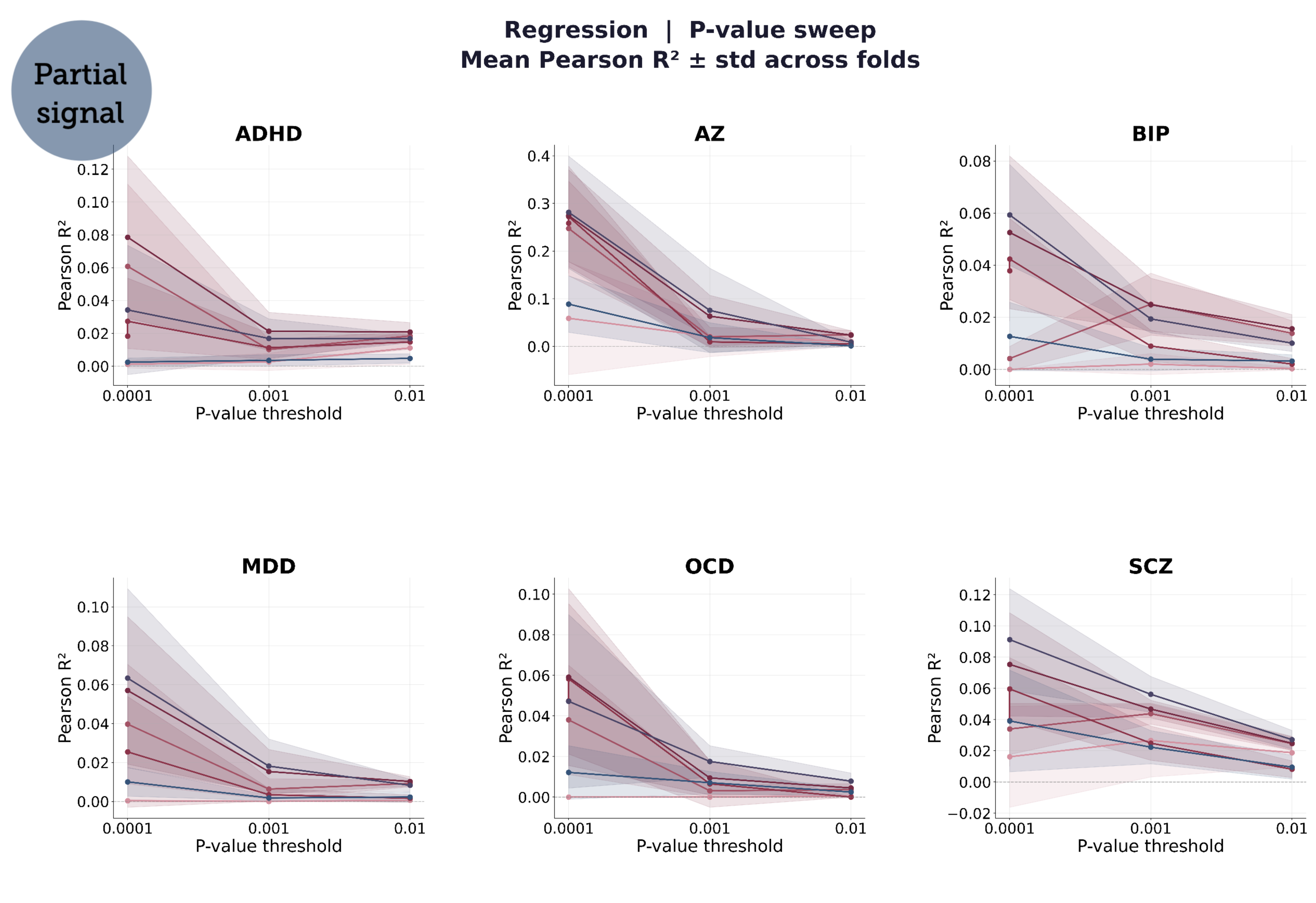
Noisy
ADHD
AZ
BIP
MDD
OCD
SCZ
Pearson R²
Lasso
Linear
Residual DNN
Ridge
TabPFN
XGBoost

Binary Classification | p = 0.001
Balanced accuracy vs row ratio



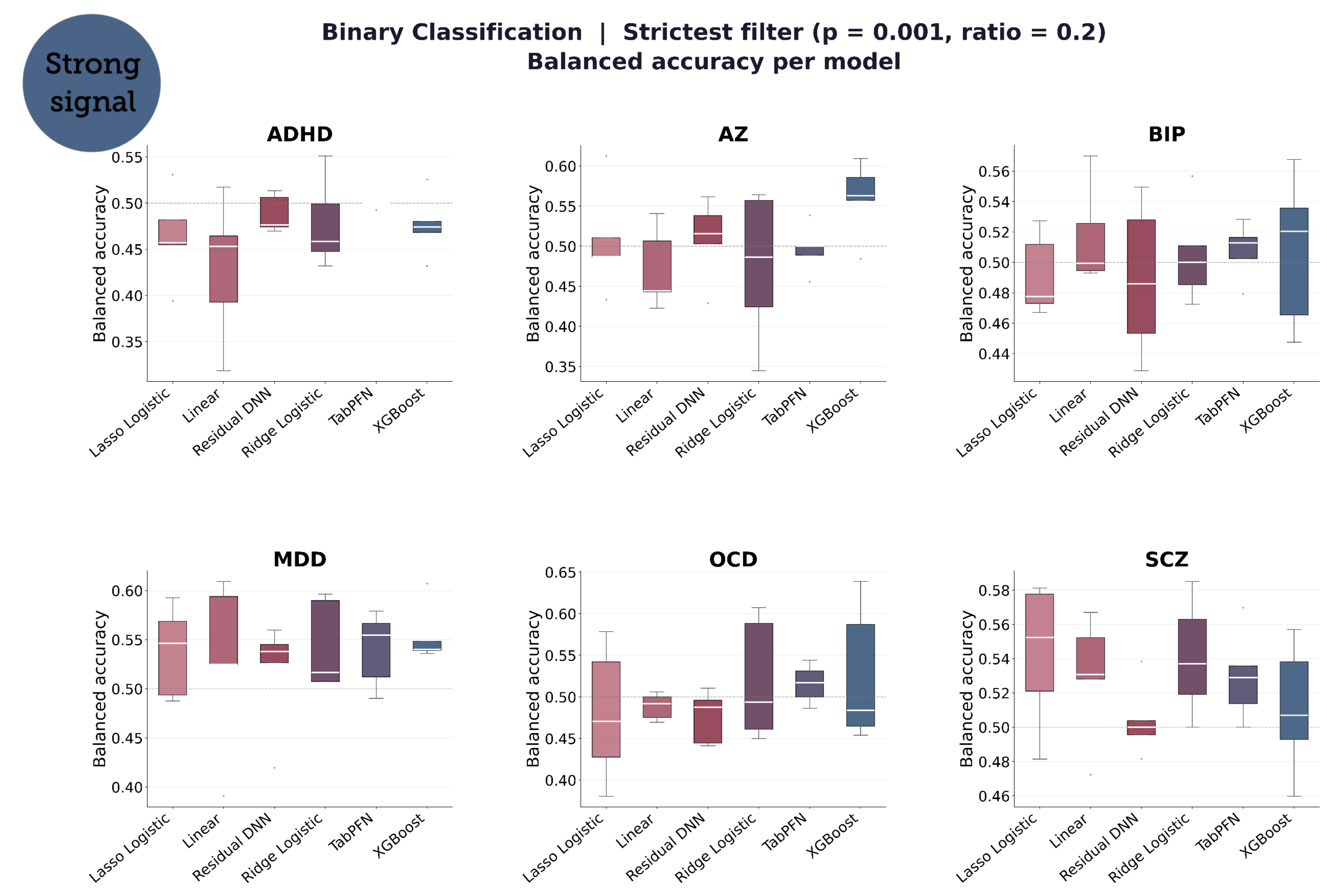
Partial signal
ADHD
AZ
BIP
MDD
OCD
SCZ
Balanced accuracy
Row ratio

Regression | P-value sweep
Mean Pearson R² ± std across folds



Partial signal
ADHD
AZ
BIP
MDD
OCD
SCZ
Pearson R²
P-value threshold

Binary Classification | Strictest filter (p = 0.001, ratio = 0.2)
Balanced accuracy per model



Strong signal
ADHD
AZ
BIP
MDD
OCD
SCZ
Balanced accuracy
Lasso Logistic
Linear
Residual DNN
Ridge Logistic
TabPFN
XGBoost